

Research article

Education and technological innovation in mitigating CO₂ emissions and fostering green economic growth in China: Marginal effects and policy threshold analyses

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ABSTRACT

This study makes a significant contribution to China's ambitious goals of achieving carbon dioxide (CO₂) neutrality and transitioning to green economic growth (GEG), and integrating the theoretical framework of the impact, population, affluence, and technology (IPAT) theory, with real-world application to reduce CO₂ and promote GEG for sustainable development. Furthermore, the study examines the ongoing theoretical debate on whether an inverted U-shaped environmental Kuznets curve (EKC) exists between technological innovations (TI) in environment-related fields and CO₂ emissions in China, using data from 1990 to 2020 and employing the threshold instrumental variable two-stage least-squares (Th-IV2SLS) model. The findings indicate that all the variables representing education contribute to reducing CO₂ emissions. The cost-effective levels of these variables to achieve CO₂ reduction are as follows: a 93% literacy rate index, 12% education expenditure as a percentage of GDP, and an average of 6 years of schooling. Furthermore, TI also contributes to CO₂ reduction, with a cost-effective level of 10.16% of TI. Educational variables promote GEG, with their respective cost-effective levels being 84% of the literacy rate index, 11.9% of education expenditure as a percentage of GDP, and an average of 5.5 years of schooling. In addition, TI promotes GEG, with a cost-effectiveness level of 10.4%. Moreover, there is a synergistic effect between education and TI that reduces CO₂ emissions; however, the synergy that promotes GEG is relatively weak. Based on these findings, policy recommendations are provided to enhance the effectiveness of education and TI in reducing CO₂ emissions and promoting GEG.

Abbreviation

AYS	Average year of schooling
CCS	Carbon capture and storage
CO ₂	Carbon emissions
EAMFP	Environmentally adjusted multifactor productivity
EDUEXPEN	Education expenditure
EKC	Environmental Kuznets curve
EMP	Employment rates
GDPPC	GDP per capita
GHG	Greenhouse gases
GEG	Green economic growth
GDL	Global data lab
IPCC	Intergovernmental panel on climate change
IPAT	Impact, population, affluence, and technology
NREC	Nonrenewable energy consumption

(continued)

OECD	Organization for economic co-operation and development
POPG	Population growth
R&D	Research and development
REC	Renewable energy consumption
STIRPAT	Stochastic impacts by regression on population, affluence, and technology
UNEP	United nations environment programme
URB	Urbanization
TI	Technological innovation
Th-IV2SLS	Threshold instrumental variable two-stage least-squares
WDI	World bank development indicators

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1. Introduction

Globally, environmental stability and eco-friendly economic practices have been subjects of intense discussion. Nations across developed, developing, and underdeveloped regions are working to identify more effective and intelligent strategies for achieving sustainability. Furthermore, the advancement of environmentally sustainable technologies is crucial for achieving the goal of ecological preservation. China's rapid economic transformation over the past few decades has established it as a global economic leader, but this growth has come with significant environmental consequences. As the world's largest emitter of greenhouse gases (GHG) which CO₂ emissions is its main responsible for global warming (Ortega-Ruiz et al., 2022), it faces immense pressure to balance its economic ambitions with sustainable practices (AlOmar et al., 2023; IPCC, 2021; UNEP, 2021). The nation's dependency on coal-fired power plants, the expansion of energy-intensive industries, and accelerated urbanization have substantially increased CO₂ emissions (Jiang and Guan, 2016; Zhang and Lin, 2012). In response, China has prioritized reducing CO₂ emissions and promoting a green, low-carbon economy as key elements of its sustainable development strategy (Ouyang and Lin, 2017). To combat climate change and achieve sustainable development, a comprehensive approach integrating education and technological innovation (TI) is necessary (Li et al., 2023; O'Neill et al., 2020).

Education is crucial for raising awareness, promoting behavioral changes, and educating individuals with the information and skills needed to meet the complex issues posed by climate change (Liu et al., 2022; Wang et al., 2021). China can foster a generation of individuals who are knowledgeable about the causes, implications, and potential solutions to climate change by including environmental education into both formal and informal curricula. Furthermore, education enables individuals to make informed decisions and adopt environmentally friendly lifestyles, hence reducing CO₂ emissions at both the individual and society levels (Jimlan, 2020; Sung, 2015).

Technological innovation provides numerous ways to reduce CO₂ emissions and promote GEG. Innovations in renewable energy sources, such as solar and wind power, and the implementation of energy-efficient technologies across various sectors can significantly reduce the CO₂ footprint of industrial and residential activities (Xu and Lin, 2017; Zhu et al., 2016; Chen and Lee, 2020). Studies have shown that education contributes to developing social responsibility within communities, thereby aiding in reducing pollution emissions (Alkhateeb et al., 2020). A nation's energy resource utilization and overall economic health are heavily influenced by its educational and human capital systems, with education having far-reaching economic effects (Balaguer and Cantavella, 2018). Education can help to mitigate the negative consequences of CO₂ emissions by fostering environmental knowledge and sustainable habits. Moreover, TI is crucial for tackling environmental issues and fostering GEG. Advances in science and technology can improve the efficiency of fossil energy usage while also expanding the possibilities for finding and exploiting new energy sources, resulting in significant CO₂ emissions reduction (Xie et al., 2021). Research on the correlation between TI and CO₂ emissions intensity can motivate governments to develop policies that maximize the benefits of scientific and TI in CO₂ reduction.

Due to ongoing scholarly investigations into finding a way to balance economic development with CO₂ emissions, "low-carbon" development has gained significant attention on a global scale (Dong et al., 2023; Yin and Yu, 2022). Technological developments provide a wide range of approaches to reduce CO₂ emissions and promote environmentally friendly economic growth. Technological advances can significantly reduce CO₂ emissions from home and industrial activities, ranging from energy-efficient technology to renewable energy sources (Xu and Lin, 2017). These advancements include energy-efficient technologies, renewable energy sources, CO₂ capture and storage, sustainable transportation, and green building design further contribute to decarbonizing

China's economy (Du et al., 2018). Most importantly, Meifeng and Lin (2021) argue that merging TI and education can strengthen each field's unique effects while also advancing integrated growth. China can accelerate its transition to a low-carbon economy, promote economic growth, and protect the environment by fostering an informed and skilled population capable of capitalizing on technological breakthroughs. The influence of TI on CO₂ emissions is increasingly evident (Zhang and Chen, 2022; C. Zhang et al., 2022), prompting ongoing scholarly investigations into the relationship (Gu et al., 2019; Hu et al., 2022; Zhu, 2022).

This study therefore aims to investigate the intricate relationships between education and innovations in environment-related technologies in mitigating CO₂ emissions and fostering GEG in China as marginal effects and policy thresholds analyses. Such investigation holds great significance within the framework of China's ambitious objectives to attain CO₂ neutrality and transition to a green economy. In addition to the fact that achieving carbon neutrality is crucial for urban ecosystem (Vais et al., 2023). Although considerable progress has been achieved in recent years, China continues to confront formidable obstacles in its efforts to decrease its CO₂ footprint and alleviate the environmental consequences of its economic operations (Jiang and Guan, 2016; Xu and Lin, 2017). Furthermore, while the importance of education and TI in promoting GEG and reducing CO₂ is widely acknowledged, there remains a gap in comprehensive knowledge regarding their marginal effects and the optimal policy thresholds necessary to maximize their influence within China's sustainable development framework. Previous research has explored the individual contributions of TI and education to reducing CO₂ and promoting sustainable practices (Amin et al., 2019; Alkhateeb et al., 2020; Chen and Lee, 2020; Hao et al., 2021; Li et al., 2021; Morgadinho et al., 2015; Xu and Lin, 2017; Zhu et al., 2016). However, a comprehensive examination that considers the combined and synergistic impacts as in this study, is essential. In addition, policymakers must establish policy thresholds for these areas to formulate effective strategies. This study is among the few that investigates such connection between TI in China and national education levels. While it distinguishes itself by employing CO₂ intensity as an analysis variable, alongside other economic factors, unlike prior studies that focused solely on CO₂. Through the identification of the policy threshold which investments in these sectors produce accelerated benefits or diminishing returns, policymakers can enhance the efficiency of resource allocation and enact measures that optimize the balance between economic growth and environmental sustainability.

Therefore, this study makes significant contributions to the existing literature. First, this study bridges the gap between theoretical framework and real-world applications, for example, verifying the validity of the theory of impact, population, affluence, and technology (IPAT) in relation to CO₂ and GEG for achieving green environment and sustainable development. Another example is the revelation by Gu et al. (2019) that an inverted U-shaped environmental Kuznets curve (EKC) could hold between TI or rather environmental innovation and CO₂ (Niu, 2021); however, almost all the existing studies are not up to date in investigating that likely feature between the relationship. Thus, this study will consider this likely nature of the relationship while estimating the relationship between TI and the CO₂ to verify this hypothesis/claim. Second, while previous studies have explored the individual roles of education and TI in promoting environmental sustainability (Zhang and Lin, 2012; Zhu et al., 2016; Xu and Lin, 2017; Cheng et al., 2021a,b), this study distinguishes itself by adopting a holistic approach. It simultaneously examines the impacts of both education and TI, as well as the synergistic effects of their interaction, in promoting environmental sustainability and GEG for sustainable development. By doing so, the finding is expected to offer a more comprehensive understanding of the intricate relationship among these factors, namely education, TI, CO₂, GEG, and their collective potential in driving sustainable development. Third, there is a scarcity of studies that determine the policy threshold levels of national education and TI, capable of mitigating the effect of

CO₂ intensity and promoting GEG. Despite these policy thresholds could have the potential to inform policy decisions in targeting CO₂ and GEG; this study has estimated such gaps. Such insights from the threshold's values, considering currently the world is collectively strives to address the pressing issue of climate change, can serve as a guide for employing education and TI as drivers for a sustainable and low-carbon future. Fifth, investigating the connection between CO₂ intensity and GEG is a new or rather the area under exploration; most of the existing studies focus on CO₂ and economic growth, thus, intensity and green aspects were neglected in such studies. Nonetheless this study prefixes to use the GEG to at least examine how purely the CO₂ intensity is connected with it for gaining facts that could aid in attaining sustainable development. Sixth, this study as contrary to most existing related studies, it makes use of national education level instead of just education enrollment or expenditure. Seventh, this study employs the threshold instrumental variable two-stage least-squares (Th-IV2SLS) model, which has not been applied before to such a relation in China despite its advancement in exploring rigor facts for environmental and economic development. This novel model is superior to other models that can execute such analysis, which includes its ability to combine multiple instrumental variables easily to obtain robust results. It also makes counting control variables easier than other models in attaining the objectives of studies (Hardy, 2016). Thus, the findings of this study could be more effective than the previous ones due to the use of advanced econometric techniques that link theory with practice in providing empirical facts.

Hence, specifically, this research work designed its objectives as follows: (1) to investigate the effects of the national education level and TI on CO₂ emissions intensity and GEG; (2) to determine the policy threshold levels of national education and TI capable of mitigating the effect of CO₂ intensity and promoting GEG; and (3) to evaluate the interaction effects between national education and TI on CO₂ intensity reduction and GEG, identifying synergies and trade-offs, while employing advanced techniques.

Practically, this study will contribute in helping policymakers promote sustainable behaviors and increase carbon efficiency. Through the development of green technology industries, this strategy not only lessens the negative effects on the environment but also promotes economic growth. Conversely, the study's academic implications include that it will serve as a foundation for incorporating educational advancements into environmental policy frameworks and that it will aid in the creation of policies that encourage skill development to support green innovation. Furthermore, it might result in more all-encompassing approaches to combating climate change via innovation and education. It also affects how inclusive education programs are created so that everyone may participate in and gain from the expansion of the green economy. Moreover, the study's conclusions can aid educational establishments in better organizing and allocating funds for initiatives linked to sustainable development.

The remainder of the study encompasses an examination of pertinent literature, a technique for accomplishing the study's objectives, the presentation and interpretation of the findings of the study, a discussion of the findings, and conclusion of the study.

2. Review of related literature

2.1. Theoretical review

The theory that connects human activities and environmental change can be traced back to the impact, population, affluence, and technology model proposed by Ehrlich and Holdren (1971). However, this model is strongly criticized, believing that the model is purely conceptual and, therefore, cannot be empirically examined. Given this fact, following Zakari et al. (2023), this study adopted the stochastic impacts by regression on population, affluence, and technology (STIRPAT) framework proposed by Dietz and Rosa (1997) due to its superiority over the traditional EKC, where income per capita is treated as an exogenous

variable and CO₂ emissions per capita is an endogenous variable. This approach ensures that the impact of population on the environment is maintained as unitary elastic (Dietz and Rosa, 1994; York et al., 2003; York et al., 2004). The essential property of the STIRPAT model was asymptotically designed to examine the impact of socio-economic changes on environmental degradation. The STIRPAT is illustrated in equation (1) as follows:

$$I_t = \alpha P_t^\beta A_t^\gamma T_t^\lambda e_t \quad (1)$$

where I_t is the environmental pressure; P_t , A_t , and T_t represent the population, affluence, and technology, respectively; β , γ , and λ are the estimated parameters, respectively; and e_t is the error term., while μ is the disturbance.

2.2. Empirical review

2.2.1. Education, CO₂ emission, and GEG

Education plays an important role in lowering CO₂ emissions by raising environmental awareness and encouraging sustainable behavior. Higher education, in particular, is linked to lower CO₂ emissions because it encourages advanced skills and knowledge that lead to innovative solutions to environmental problems (Eyuboglu and Uzar, 2021). Secondary education in Saudi Arabia has been found to be reducing the negative impact of CO₂ emissions, thus improving environmental quality (Alkhateeb et al., 2020). The relationship between economic growth and CO₂ emissions is complex. While early stages of economic growth frequently result in increasing emissions, the relationship can evolve into an inverted U-shape EKC, with higher income levels eventually leading to a drop in emissions (Narayan et al., 2016). According to studies, energy use is a major source of CO₂ emissions. However, investments in renewable energy and TI can help to reduce these effects (Ito, 2017). Moreover, GEG aims to decouple economic development from environmental degradation. Investments in research and development (R&D) and green technologies are crucial for reducing CO₂ emissions while promoting economic growth (Fernández et al., 2018). Government spending on education and green investments can significantly contribute to green economic growth by fostering a more educated workforce and promoting sustainable practices (Xu et al., 2023).

Human capital development has been shown to have an adverse effect on energy consumption, thereby enhancing environmental quality (Akram et al., 2019; Shahbaz et al., 2019). Balaguer and Cantavella (2018) argued that, a nation's energy resources are highly dependent on its educational and human capital systems, and that education can have far-reaching effects on the economy. The authors also postulated that educational initiatives could counterbalance the adverse consequences of CO₂ emissions. According to the findings of Goetz et al. (1998), the USA maintains favorable environmental conditions, predominantly as a result of the income and structural transformations it fosters, and its highly educated populace. Furthermore, according to Wu (2017), human capital positively influences CO₂ emissions. Similarly, Bano et al. (2018) investigated the correlation between human capital and CO₂ emissions in the context of Pakistan. Their findings indicated that long-term advancements in human capital led to a reduction in CO₂ emissions. Nevertheless, no relationship can be observed between these two variables in the intermediate term.

Li and Ouyang (2019) found that an elevated degree of human capital concurrently elevates immediate CO₂ emissions and diminishes long-term CO₂ emissions. Moreover, Ahmed and Wang (2019) looked at the relationship between CO₂ emissions and human capital in the case of India. According to their research, a higher level of human capital reduces ecological footprint by improving short- and long-term environmental consequences. In a similar vein, Yao et al. (2019) found that human capital has a negative impact on CO₂ emissions, while Dedeoglu et al. (2021) concluded that human capital has no discernible impact on

CO₂ emissions. According to Alkhateeb et al. (2020), education can play a significant role in lowering pollution emissions by encouraging social responsibility within a society. A negative correlation was established between education and CO₂ emissions by Liu et al. (2022). Li and Ullah, (2022) found that in BRICS economies, CO₂ emissions are substantially reduced with increasing levels of education, whereas a decline in educational attainment has led to an amplification of CO₂ emissions. Zafar et al. (2022) found strong correlations between environmental deprivation and education. On the contrary, Zhang et al. (2022) revealed a decline in environmental quality due to the tendency for education to contribute to increased emissions. In their study examining the effects of education on CO₂ emissions in China, Xin et al. (2023) discovered that indicators of human capital, including the average year of schooling and the literacy rate, have a long-term mitigating effect on CO₂ emissions. Based on the nature of the contradictory results, it can be deduced that the findings regarding the relationship between human capital, or education, and CO₂ emissions remain a matter of contention. As a result, there is additional space for research.

2.2.2. Technological innovation, CO₂ emissions, and GEG

TI is critical in managing CO₂ emissions and GHG impacts. The relationship between technological progress and environmental impact is complex, covering improvements in energy efficiency, the development of low-carbon technology, and changes in industrial practices. Several research have investigated this dynamic, indicating both positive outcomes and ongoing challenges. Research shows that green innovation has a major impact on corporate sustainability, particularly in energy-intensive industries (Lou et al., 2022). In China, for example, green product innovation, recycling, and green PR have been shown to improve financial, environmental, and social performance, thereby contributing to a low-carbon circular economy. Similarly, a study on the impact of green innovation in Chinese manufacturing emphasizes the necessity of energy conservation and emission reduction efforts in meeting climate change mitigation targets (Li et al., 2020). The introduction of CO₂ trading schemes, such as California's cap-and-trade program, aims to reduce GHG emissions by setting limits and allowing the trading of emission permits. However, evidence suggests that these programs can result in social disparities, with economically disadvantaged neighborhoods experiencing higher local GHG and co-pollutant emissions, indicating a need for more equitable policy implementations (Cushing et al., 2018). Furthermore, technological options such as CO₂ capture and storage (CCS) have been proposed for directly reducing industrial CO₂ emissions. Public support for such technologies is critical, but research shows that effective communication tactics, including the use of social norms, are required to increase public acceptability and foster widespread adoption (Wang et al., 2021). Persuasive technology shows promise for attitude adjustment but incentives' relationship with low-carbon behavior complex (Thanapongporn et al., 2023). In addition to TI, changes in customer behavior and lifestyle choices are crucial. For example, dietary CO₂ emissions have been connected to oxidative and inflammatory status in humans, emphasizing the larger health consequences of sustainable diets (Monserrat-Mesquida et al., 2023). Furthermore, the environmental impact of motorized transport, which is influenced by factors such as automobile ownership and trip distance, emphasizes the need for integrated methods that combine technology advances with legislative measures to minimize emissions (Brand et al., 2013). Despite advancements, some challenges remain. The perception of risk associated with new technologies, such as CO₂ removal methods, can hinder their acceptance and deployment. Public mistrust, fueled by past experiences with controversial technologies like fracking, can negatively impact perceptions of CO₂ removal efforts, suggesting a need for transparent and trustworthy communication (Cox et al., 2022).

In addition to these findings, Grossman and Krueger (1995) contend that TI are the primary cause of emission reduction. TI, specifically those pertaining to environmental technology, have had a profound impact on the environment, economy, and society at large. Lee and Mo (2011), for

instance, argued that environmental innovation in biofuel is associated with the reduction of CO₂ emissions but has no bearing on the economy. In a similar vein, Carrion-Flores and Innes (2010) found that toxic contamination in the US has decreased significantly as a result of environmental innovation and stricter policy. However, CO₂ emissions may not be significantly reduced through the implementation of environmental innovation if the time horizon is limited. Therefore, the advantages of implementing environmental innovation are contingent upon their long-term implementation (Mongo et al., 2021). Furthermore, the efficacy of CO₂ emission reduction is greater for environmental innovation manifested in renewable energy as opposed to fossil fuel energy (Wang and Zhu, 2020). The findings of Dauda et al. (2021) are comparable when applied to Africa. Moreover, an inverted U-shaped EKC may exist between environmental innovation and CO₂ emissions, according to Gu et al. (2019), which consists of an initial increase in CO₂ emissions followed by reductions (Niu, 2021). Besides, according to Gorg and Greenaway (2004), the diffusion of foreign technology contributes positively to CO₂ reduction. According to the findings of Zhang et al. (2014), industrial upgrading would decrease CO₂ emissions. In addition, Zhang et al. (2018) demonstrated that enterprise agglomeration enhancement is one of the most important methods for reducing industrial CO₂ emissions. Hao et al. (2021) found that the effect of environmental regulation on CO₂ emissions is complementary. CO₂ emissions are increased as a result of environmental innovation in the form of nonrenewable energy innovation (Mensah et al., 2018). Although the degree of environmental innovation is also significant, CO₂ emissions are likely to rise in low-income, middle-income, and middle-income countries with low technology and low CO₂ emissions, notwithstanding the implementation of environmental innovation (Chen and Lee, 2020). Hence, nations experiencing rapid economic expansion may face a disadvantage, given that the immediate effects of environmental innovation might not materialize in the form of CO₂ reductions (Yu and Du, 2019). In the presence of high-income inequality, Bai et al. (2020) contend that innovation in renewable energy and fossil fuel energy may contribute to an increase in CO₂ emissions.

2.3. Literature gaps

From the foregoing of both the theoretical and empirical literature, it can be observed that prior researches (Xu and Lin, 2017; Zhang and Lin, 2012; Zhu et al., 2016) have investigated the distinct contributions of education and TI in advancing environmental sustainability, but no study examined the combined effects of education and TI on CO₂ emissions in the pursuit of sustainable development and GEG. Furthermore, there is contradictory evidence in the findings, probably due to the reason of scarce studies that centered on precisely such relations. In addition, government environmental regulations and industrial structure make up the bulk of the literature on CO₂ emissions. Moreover, little study has been done to determine the policy threshold levels for national education and TI that can promote GEG and lessen the effects of rising CO₂ emissions. Such benchmarks might serve as a guide for decision-makers as they work to reduce CO₂ emissions and boost the economy. Additionally, it is evident that the utilization of CO₂ emissions intensity in this study is restricted to total CO₂ emissions and education is not referred to as the national education level in this study. Similarly, Gu et al. (2019) identified the possibility of an inverted U-shaped EKC between TI, specific innovations in environment-related technologies, and CO₂ emissions. Specifically, the EKC depicts an initial increase in CO₂ emissions followed by reductions (Niu, 2021).

3. Methodology

3.1. Data and sources

The focus variables are national education level (*EDU*) in which, following Xin et al. (2023), a composition of three different factors were

considered: literacy rate (*LITERACY*) as % of people ages 15 and above and this is in accordance with the UNESCO definition in which 15 to 24 is considered youth literacy rate while the elderly to ages 65 and above, education expenditure (*EDUEXPEN*) as total government spending on education (% of GDP), and year of schooling measured as average year of schooling (*AYS*); technological innovation (innovations in environment-related technologies as % of total technological innovation) (*TI*); carbon emissions intensity (CO_2 emissions per capita in metric tons) (CO_2); and green economic growth (*GEG*) which indicates whether economic growth is becoming greener with more efficient use of natural capital, in other words, it measures advancements concerning the sustainable and greener economy.

For this variable of *GEG*, following Kasztelan et al. (2020) and Hussain et al. (2022), the OECD-developed environmentally adjusted multifactor productivity index (EAMFP) (Rodríguez et al., 2018) will be implemented. Natural capital inputs and undesirable outputs, i.e., negative byproducts of the production process, are accounted for in EAMFP. For a given increase in input employment, it rises in tandem with a rise in GDP or a decline in pollution (Kasztelan et al., 2020). The EAMFP growth represents the remainder, or the proportion of GDP growth adjusted for pollution, that cannot be accounted for by increases in the utilization of factor inputs, such as labour, produced capital, and natural capital. Consequently, the EAMFP establishes an explicit correlation between "green" and "growth" in order to generate a metric that evaluates both economic and environmental performance.

The control variables include research and development expenditure (% of GDP) (*R&D*), renewable energy consumption (% of total energy consumption) (*REC*), nonrenewable energy consumption (% of total energy consumption) (*NREC*), international trade (*TRADE*), urbanization (*URB*), population growth (*POPG*), GDP per capita (*GDPPC*), inflation rate (*INF*), and employment rates (*EMP*). The data for these variables spans from 1990 to 2020, chosen due to the availability of data for the focus variables, including *GEG* and *TI*. The variables were sourced from the Organization for Economic Cooperation and Development (OECD) (2023), World Bank Development Indicators (WDI) (2023), and Global Data Lab (GDL) (2023). However, the tabular presentation of the variables, their measures, codes, and sources are shown in Table 1.

3.2. Functional specifications of the study

The functional specifications of the study adopted and modified from

Table 1
Variables, measures, codes, and sources.

Variables	Measures	Codes	Sources
National education level			
-Literacy rate	Index in % of 100	LITERACY	WDI
-Education expenditure	% of GDP	EDUEXPEN	WDI
-Year of schooling	Average year of schooling	AYS	GDL
Technological innovation	% of total	TI	OECD
CO_2 emissions intensity	Metric tons	CO_2	WDI
Green economic growth	Index of EAMFP	GEG	OECD
Research and development expenditure	% of GDP	R&D	WDI
Renewable energy consumption	% of total energy	REC	WDI
Nonrenewable energy consumption	% of total energy	NREC	WDI
International trade	% of GDP	TRADE	WDI
Urbanization	% of total population	URB	WDI
Population growth	Annual %	POPG	WDI
GDP per capita	RGDP per capita in LCU	GDPPC	WDI
Inflation rate	Annual %	INF	WDI
Employment rate	Rate (modeled ILO estimate)	EMP	WDI

Hussain et al. (2022). According to the specific objectives of this study as identified under the introduction section, four specifications are going to be made where equations (2) and (3) are for objectives one and two, equations (4) and (5) are for objective three. These equations are stated as follows:

$$\text{CO}_2 = \beta_0 + \beta_1 \text{EDU} + \beta_2 \text{TI} + \beta_3 \text{TI}^2 + \beta_4 \text{R\&D} + \beta_5 \text{NREC} + \beta_6 \text{REC} + \beta_7 \text{TRADE} + \beta_8 \text{URB} + \beta_9 \text{POPG} + \beta_{10} \text{GDPPC} + \beta_{11} \text{INF} + \beta_{12} \text{EMP} + \mu_t \quad (2)$$

$$\text{GEG} = \beta_0 + \beta_1 \text{EDU} + \beta_2 \text{TI} + \beta_3 \text{R\&D} + \beta_4 \text{NREC} + \beta_5 \text{REC} + \beta_6 \text{TRADE} + \beta_7 \text{URB} + \beta_8 \text{POPG} + \beta_9 \text{GDPPC} + \beta_{10} \text{INF} + \beta_{11} \text{EMP} + \mu_t \quad (3)$$

$$\text{CO}_2 = \beta_0 + \beta_1 \text{EDU} * \text{TI} + \beta_2 \text{EDU} * \text{TI}^2 + \beta_3 \text{R\&D} + \beta_4 \text{NREC} + \beta_5 \text{REC} + \beta_6 \text{TRADE} + \beta_7 \text{URB} + \beta_8 \text{POPG} + \beta_9 \text{GDPPC} + \beta_{10} \text{INF} + \beta_{11} \text{EMP} + \mu_t \quad (4)$$

$$\text{GEG} = \beta_0 + \beta_1 \text{EDU} * \text{TI} + \beta_3 \text{R\&D} + \beta_4 \text{NREC} + \beta_5 \text{REC} + \beta_6 \text{TRADE} + \beta_7 \text{URB} + \beta_8 \text{POPG} + \beta_9 \text{GDPPC} + \beta_{10} \text{INF} + \beta_{11} \text{EMP} + \mu_t \quad (5)$$

where β_0 is the constant term; $\beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6, \beta_7, \beta_8, \beta_9, \beta_{10}, \beta_{11}$, and β_{12} are the respective coefficients of the variables, namely national education level (*EDU*), technological innovation (innovations in environment-related technologies) (*TI*), CO_2 emissions intensity (CO_2), green economic growth (*GEG*), research and development (*R&D*), renewable energy consumption (*REC*), nonrenewable energy consumption (*NREC*), international trade (*TRADE*), urbanization (*URB*), population growth (*POPG*), GDP per capita (*GDPPC*), inflation rate (*INF*), and employment rates (*EMP*), respectively: μ_t is the stochastic term of the estimate; and the * indicates interaction term. This shows that five different models, as specified in equation (2) through 5, are to be estimated. Furthermore, it can be observed that the control variables in the specified models have a cross appearance, i.e., they appeared in multiple equations, thus, likely to have some difference in magnitude or significance; they will be interpreted as they are, and this will not affect the focus results of the study since they are not the target variables or the purpose variables of the study. In fact, no two or more equations contain the same focus variable but control variables. Moreover, following the theory of the EKC, this study hypothesized the quadratic effects of *TI* on CO_2 emissions, considering the view of Gu et al. (2019) that an inverted U-shaped EKC could hold between *TI* and CO_2 emissions.

3.3. Estimation techniques

The estimation techniques employed to achieve the objectives of this research were presented as follows according to the order they were executed. First, is the graphical illustration of the variables under the study for trend observation. Second, is the descriptive statistics to display the statistical characteristics of the variables under the study. Third, is the conduct to examine the stochastic properties of the variables under the study using Augmented Dickey-fuller (ADF), Phillips-perron (PP), and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) unit root tests so as to ensure no variable is integrated of order two, i.e., $I(2)$, which deemed not fit for the analysis. Fourth, is the estimation of the (Th-IV2SLS) model, which is the model employed to estimate the relationships specified in equations (2)–(5). The choice of the Th-IV2SLS model is due to its superior handling of threshold estimates and ability to manage multiple instrumental variables, making it more flexible and robust than other models for such analyses. This model excels in incorporating control variables effectively, thereby mitigating issues related to omitted variable bias and measurement errors (Hardy, 2016). Studies highlight its performance advantage in the presence of endogenous variables, providing more accurate and unbiased estimates compared to traditional methods like maximum likelihood estimator and standard 2SLS (Hayakawa and Sun, 2019; Maydeu-Olivares et al., 2019; Nestler, 2015). Furthermore, it maintains accuracy even when the model is misspecified; hence, robust and reliable (Nestler, 2014).

Th-IV2SLS is a threshold regression model that estimates a linear regression model of the dependent variable (the second stage) using estimated values of the problematic predictors (the first stage) that are uncorrelated with the error terms using instrumental variables (the threshold II2SLS model). As the computed values are therefore dependent on variables that are not correlated with the errors, the Th-IV2SLS model yields optimal results. The estimation phase of this model will commence by investigating the statistical significance of the threshold effects using the following hypotheses.

H0. $\beta_1 = \beta_2$ there is no significant threshold value.

H1. $\beta_1 \neq \beta_2$ there is significant threshold value.

The presence of a threshold in the estimation is confirmed if the null hypothesis of no threshold was rejected by the linear regression ($\beta_1 = \beta_2$). Consequently, upon rejecting the null hypothesis, the presence of the alternative hypothesis ($\beta_1 \neq \beta_2$) indicates the existence of a threshold (γ) in the estimation. The threshold value is reached, however, when the significance level of the bootstrap p -value indicates that the null hypothesis has been rejected. By employing a confidence level of 95 percent, the model establishes the Hansen critical value (2000, p.582).

In the subsequent stage of this model, an output method based on a sample-splitting technique is implemented. The sample is initially divided into two subsamples according to the threshold value. Furthermore, the bootstrapping method is employed by the threshold model, which promotes the generation of confidence level regions through the utilization of 1000 repetitions for every regression. Indicated will be the bootstrap p -value and the corresponding 95% confidence interval. As soon as the threshold value estimate (p -value and likelihood ratio) and threshold testing produce evidence of the threshold, we can commence the process of regressing the estimates in order to derive the parameter estimates.

By utilizing the threshold, the sample is divided into two regimes: low and high. As mandated by the second objective of the study, TI and national education level will function as threshold variables in their respective models. Consequently, these factors will be incorporated into both the explanatory and threshold variables. Both Bose et al. (2008) and Alfada (2019) employed a comparable methodology and did not observe any statistical issues. The expression for the equation is presented in equations (6) and (7) as follows:

$$CO_{t2} = \beta_1 X_t I(EDU_t; TI_t \leq \gamma) + \beta_2 X_t I(EDU_t; TI_t > \gamma) + e_t \quad (6)$$

$$GEG_t = \beta_1 X_t I(EDU_t, TI_t \leq \gamma) + \beta_2 X_t I(EDU_t, TI_t > \gamma) + e_t \quad (7)$$

where EDU and TI are the threshold variables; γ is the threshold value; $I(\cdot)$ is an indicator function of the threshold variable; the subscript t denotes the period; X_t consists of the focus and controls explanatory variables; and e_t is the stochastic term.

4. Results presentation and interpretation

The analysis started with the graphical representation of the variables under study to understand the nature of their trend, as illustrated in Fig. 1. These variables include *LITERACY*, *EDUEXPEN*, *AYS*, *TI*, *CO₂* emissions intensity, *GEG*, *R&D*, *REC*, *NREC*, *TRADE*, *POPG*, *URB*, *GDPPC*, *INF*, and *EMP* over the horizon of 1990–2020. In line with the figure, literacy rate and average year of schooling portray an upward trend, thus, a rising trend throughout the period. However, education expenditure portrays a fluctuating downward and upward trend, in which right from the beginning, it continues to decline up to around 2007, then it picked up and continued to rise to around 2013, then it again continued to fall up to the end of the remaining period. Technological innovation, i.e., innovations in environment-related technologies, show a base-shape trend where it initially continued to fall with little fluctuations until 2000, then it picked up and continued to increase up to the end of the period, remaining though with little fluctuations. *CO₂*

emissions intensity displays an upward trend throughout the period, though with gentle fluctuations. *GEG* shows a fluctuated trend that is intricate between downward and upward, where it initially increases up to around 1993, then declines to around 1998, then rises to up to 2000, then again continues to decline up to around 2015, then it picked up and continue to rise to the end of the remaining period. *R&D* expenditure is increased throughout the period, which indicates an upward trend. Renewable energy consumption portrays a partial u-shape trend, initially starting to decline up to around 2011, then picking up and continuing to rise up to the end of the remaining period. Nonrenewable energy consumption depicted a small letter n-shape trend, which initially started to rise around 2011 and then fell to the end of the remaining period. International trade shows a small letter n-shape trend though with little fluctuations, where it initially rises to up to 2006, then continues to fall gently and also in a fluctuated way up to the end of the remaining period. Population growth shows a downward trend throughout the period. Urbanization and GDP per capita display an upward trend throughout the period. Inflation depicted a behavior almost similar to that of green economic growth, thus, a fluctuating trend and is intricate to directly say fluctuated downward or upward, where it initially rises to 1994, then continues to fall to up 1999, then rises a little to 2001, then declines in 2002, then continues to increase gently in a fluctuated manner up to 2008, then it falls in 2009, then rises to 2011, again falls and continues to decline up to the end of the remaining period. The employment rate declined throughout the period, thus creating a downward trend.

Table 2 presents the descriptive statistics of the variables to help understand the characteristics of the variables under study. From the table, it can be observed that the respective means of all the variables are higher than their corresponding standard deviation except in the case of *GEG* and *INF*. This suggests that their observations are tightly centered around their mean except for those of *GEG* and *INF*, which are more spread out. Furthermore, *GDPPC* is the variable with the highest standard deviation, which indicates that these variable exhibits higher variability than the other variables. At the same time, *POPG* is the variable with the least standard deviation, which means that it exhibits lower variability than the other variables. Moreover, observing the skewness distribution, it can be seen that all the variables have positive skewness except *LITERACY*, *TI*, *R&D*, *NREC*, and *EMP*, which have negative skewness.

Table 3a, 3b and 3c reports the ADF, PP, and KPSS unit root tests in which, according to the tables, the variables are stationary at least at integrated order one, i.e., $I(1)$ or better, i.e., at integrated order zero. Therefore, no variable is $I(2)$; thus, none of the variables can be discarded from the analysis.

Table 4 displays the threshold instrumental variables (2SLS) regression results for model 1, which is to investigate the effects of national education level and technological innovation on *CO₂* emissions intensity in China. From the table, national education (*EDU*) variables, i.e., *LITERACY*, *EDUEXPEN*, and *AYS* are found to be negatively related with the *CO₂*. The impacts are significant except that of *EDUEXPEN* which is insignificant, thus, minimal even though its impacting. The impact direction is inline of the a priori expectation, where a 1% increase in literacy rate, 1% increase in education expenditure as a percentage of GDP, and 1 year increase in average year of schooling will lead to a reduction in the release of *CO₂* emission intensity by .231, .013, and 6.31 metric ton of *CO₂*, respectively. However, the levels of national education variables effective in reducing the release of the emissions are 93% literacy rate, 12% education expenditure as a percentage of GDP, and 6years average year of schooling. This means that national education through literacy rate, education expenditure, and average year of schooling are aiding in achieving the goal for lowering the release of *CO₂* emissions intensity in China and the cost-effective levels to reduce the release of the emissions are 93% literacy rate, 12% education expenditure as a percentage of GDP, and 6years average year of schooling. Furthermore, *TI* and *TI²* both significantly impact *CO₂* positively and

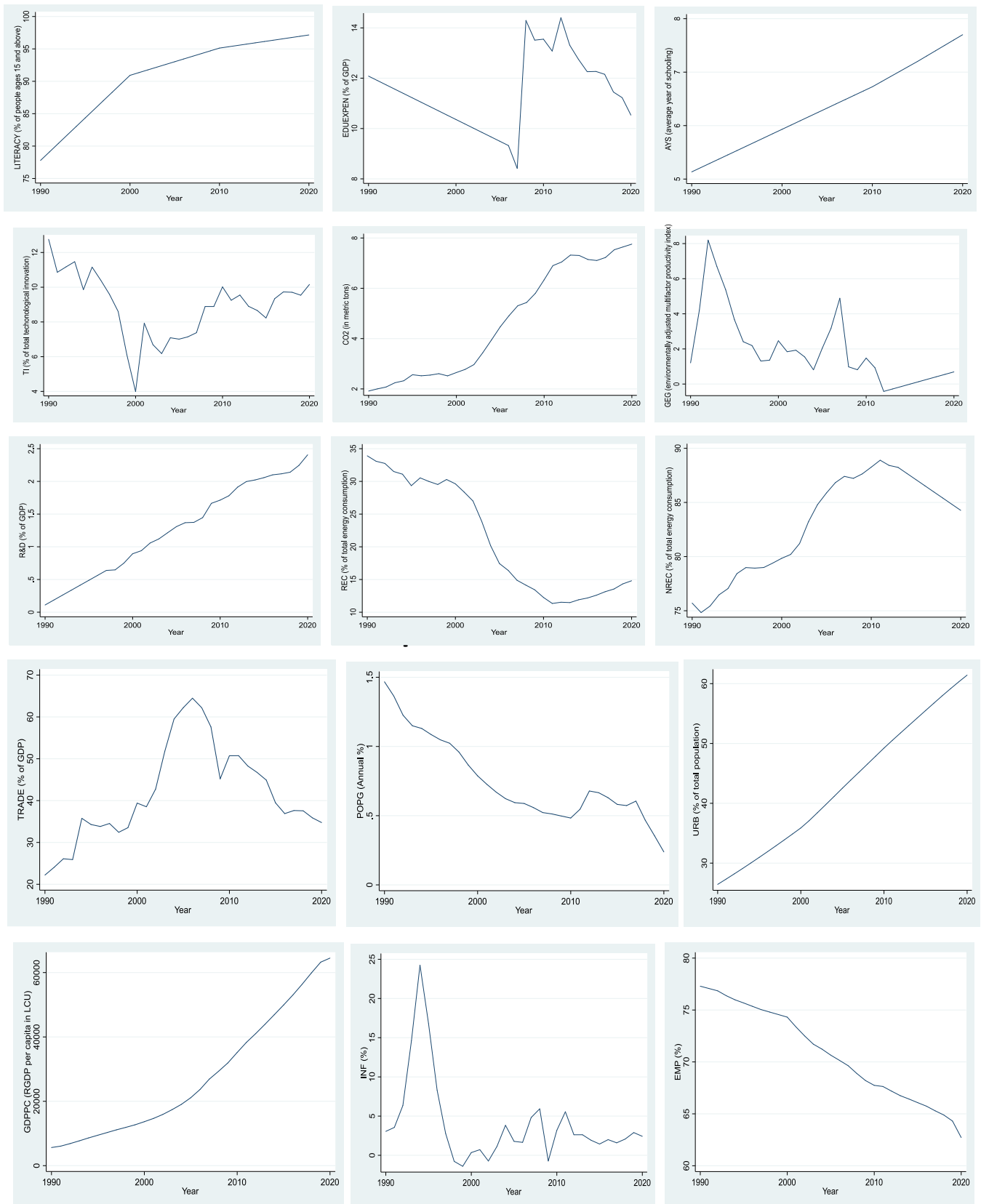


Fig. 1. Graphical representation of the variables Source: Author's Computation.

Table 2
Descriptive statistics of the variables.

	LITERACY	EDUEXPEN	AYS	TI	CO ₂	GEG	R_D
Mean	91.05128	11.45783	6.357484	8.911613	4.655473	1.958788	1.266837
Median	93.02235	11.39400	6.328000	9.250000	4.467696	1.360492	1.307920
Maximum	97.15000	14.40717	7.700000	12.76000	7.756138	8.199227	2.406660
Minimum	77.78506	8.415130	5.134001	3.990000	1.914546	−.419840	.110300
Std. Dev.	5.825225	1.493079	.765169	1.848908	2.181781	2.073024	.703161
Skewness	−.938125	.172094	.111182	−.442071	.148304	1.388579	−.063029
Kurtosis	2.618695	2.420576	1.862166	3.266486	1.335124	4.460409	1.687733
Sum	2822.590	355.1926	197.0820	276.2600	144.3197	60.72243	39.27194
Sum Sq. Dev.	1017.997	66.87856	17.56451	102.5538	142.8050	128.9228	14.83308

	REC	NREC	TRADE	POPG	URB	GDPPC	INF	EMP
Mean	21.17645	83.03421	41.60826	.749184	43.02432	27,629.04	4.018973	70.78852
Median	17.44000	84.79646	38.52736	.630326	42.52200	21,115.65	2.619524	70.65200
Maximum	33.91000	88.89836	64.47888	1.467303	61.42800	64,505.39	24.25699	77.29500
Minimum	11.34000	74.83197	22.19947	.238041	26.44200	5636.080	−1.401473	62.71900
Std. Dev.	8.550465	4.596078	11.52773	.303240	11.03162	19,057.55	5.453344	4.443487
Skewness	.193545	−.368418	.409601	.740091	.115290	.587606	2.293084	−.057521
Kurtosis	1.267064	1.653499	2.360943	2.675779	1.702105	1.973879	8.191219	1.653520
Sum	656.4700	2574.061	1289.856	23.22469	1333.754	856,500.3	124.5882	2194.444
Sum Sq. Dev.	2193.313	633.7180	3986.654	2.758629	3650.901	1.09E+10	892.1688	592.3373

Source: Author's computation.

Table 3a
Unit root test results (ADF).

Null Hypothesis: the variable has a unit root									
At Level									
		LITERACY	EDUEXPEN	AYS	TI	CO ₂	GEG	R_D	
With Constant	t-Statistic	-3.0275**	-2.2446	2.0839	-2.5724	-.5591	-1.9017	-.2950	
With Constant & Trend	t-Statistic	-2.5974	-2.3082	-.0858	-2.3034	-2.1934	-3.4634*	-2.3095	
Without Constant & Trend	t-Statistic	.1516	-.5066	2.1455	-.8557	1.1213	-1.3960	5.9193	
At First Difference									
		d(LITERACY)	d(EDUEXPEN)	d(AYS)	d(TI)	d(CO ₂)	d(GEG)	d(R_D)	
With Constant	t-Statistic	-1.2187	-6.6794***	.2393*	-6.5701***	-2.4951	-4.8879***	-4.6647***	
With Constant & Trend	t-Statistic	-1.8657	-6.5530***	-1.8531	-6.7366***	-2.4177	-4.7535***	-4.5433***	
Without Constant & Trend	t-Statistic	-1.7403*	-6.7901***	1.2529	-6.6938***	-1.6935*	-4.9508***	-.5034	
At Level									
		REC	NREC	TRADE	POPG	URB	GDPPC	INF	EMP
With Constant	t-Statistic	-1.3883	-2.8474*	-1.7754	-1.0038	-.6606	-1.3602	-3.0091*	1.1051
With Constant & Trend	t-Statistic	-1.0226	-.3875	-.8862	-2.1767	-3.4781*	-2.1837	-3.0984	-2.1607
Without Constant & Trend	t-Statistic	-1.5218	.8647	.0611	-1.9150*	-.5714	-1.6768*	-1.6098	-1.5960
At First Difference									
		d(REC)	d(NREC)	d(TRADE)	d(POPG)	d(URB)	d(GDPPC)	d(INF)	d(EMP)
With Constant	t-Statistic	-2.4069	-2.3166	-4.0941***	-2.7362*	-1.8512	-1.6031	-3.4965*	-1.5180
With Constant & Trend	t-Statistic	-2.6291	-3.7178**	-4.5080***	-2.5174	-.9692	-.4439	-3.4308*	-1.7299*
Without Constant & Trend	t-Statistic	-2.2032**	-2.0842*8	-4.1521***	-2.1501**	.0940	-.4573	-3.5227**8	.2423*

negatively, respectively, but TI^2 is in line with a priori expectation, which portrays a negative relationship between TI and CO_2 . It shows that a 1% quadratic increase in TI , will lead a reduction in CO_2 emission intensity by .009 metric ton of CO_2 and the level of TI on innovations in environment related technologies effective to reduce the release in of the emissions is 10.16% of all technologies. This suggests that TI on innovations in environment related technologies are lowering the release of CO_2 emissions intensity in China and the cost-effective level to reduce the release of the emissions is 10.16% of innovations in environment related technologies as a percentage of all technologies.

Other variables, namely $R\&D$, REC , $POPG$, URB , $GDPPC$, INF , and EMP are negatively related to the CO_2 , while $NREC$, $TRADE$, and $GDPPC$ are positively related to it. All are significant except the $POPG$ and $NREC$; thus, their impacts are minimal even though they satisfied the a priori expectations that an increase in $NREC$ and an increase in the size of the population is promoting the release of more CO_2 emissions. It ascertains that a 1% increase in $R\&D$ as a percentage of GDP, 1% increase in REC as a percentage of total energy consumption, 1% increase in $POPG$, URB as a percentage of the total population, $GDPPC$, INF , and EMP will lead to

reduction in the release of CO_2 emissions intensity by 2.216, .117, .261, .321, .009, and .192 metric ton of CO_2 , respectively, and 1% increase in $NREC$ as a percentage of total energy consumption, 1% increase in $TRADE$ as a percentage of GDP, and 1% increase in $GDPPC$ will lead to an increase in the release of CO_2 emissions intensity by .029, .018, and .003 metric ton of CO_2 , respectively.

Fig. 2 shows the plot of actual vs. predicted for model 1. It can be observed from the figure that the variables used to estimate the effect of education and technological innovation on CO_2 emissions intensity and the policy threshold levels of national education and technological innovation capable of mitigating the effect of CO_2 emissions intensity can potentially predict the model.

Table 5 shows the threshold instrumental variables (2SLS) regression results for model 2, which investigates the effects of national education level and technological innovation on green economic growth in China. According to the table, there is a positive relationship between GEG and national education variables, i.e., $LITERACY$, $EDUEXPEN$, and AYS and GEG and the impacts are significant except that of $LITERACY$, where a 1% increase in literacy rate, 1% increase in education expenditure as a

Table 3b
Unit root test results (PP).

Null Hypothesis: the variable has a unit root									
At Level									
		LITERACY	EDUEXPEN	AYS	TI	CO ₂	GEG	R_D	
With Constant	t-Statistic	−7.9645***	−2.1746	4.5786	−2.5399	−.0967	−2.0903	−.2873	
With Constant & Trend	t-Statistic	−2.1770	−2.3082	2.4301	−2.2082	−1.7970	−3.6103**	−2.4551	
Without Constant & Trend	t-Statistic	3.1267	−.5077	48.0103	−.8905	2.6150	−1.3788	3.7908	
At First Difference									
		d(LITERACY)	d(EDUEXPEN)	d(AYS)	d(TI)	d(CO ₂)	d(GEG)	d(R_D)	
With Constant	t-Statistic	−1.1980	−6.7096***	−.1139	−6.5905***	−2.4951	−7.5318***	−4.5599***	
With Constant & Trend	t-Statistic	−1.9335	−6.5806***	−1.7928*	−6.8965***	−2.4177	−9.1211***	−4.4093***	
Without Constant & Trend	t-Statistic	−1.7979*	−6.8223***	1.5019*	−6.7166***	−1.6935*	−5.9012***	−1.2047	
At Level									
		REC	NREC	TRADE	POPG	URB	GDPPC	INF	EMP
With Constant	t-Statistic	−1.3165	−1.6166	−1.8026	−1.9296	2.0058	4.1895	−2.2934	1.3840
With Constant & Trend	t-Statistic	−.7592	.4052	−1.0272	−2.1675	−2.9285	−1.4271	−2.5713	−1.7145
Without Constant & Trend	t-Statistic	−2.2164**	1.1155	−.0768	−3.9631***	13.7038	8.6445	−1.8322*	−7.2569***
At First Difference									
		d(REC)	d(NREC)	d(TRADE)	d(POPG)	d(URB)	d(GDPPC)	d(INF)	d(EMP)
With Constant	t-Statistic	−2.4486	−2.4713	−4.0980***	−2.6290*	−1.7442	−1.6371	−3.6266**	−1.4847
With Constant & Trend	t-Statistic	−2.6678	−3.7211**	−4.5073***	−2.3730	−.4250	−.7280	−3.5119*	−1.8936
Without Constant & Trend	t-Statistic	−2.1646**	−2.1924**	−4.1542***	−1.9732**	2.4086**	−2.4573**	−3.7349***	.5960

Table 3c
Unit root test results (KPSS).

Null Hypothesis: the variable is stationary									
At Level									
		LITERACY	EDUEXPEN	AYS	TI	CO ₂	GEG	R_D	
With Constant	t-Statistic	.6751**	.1812	.7340**	.1733	.6956**	.5095**	.7308**	
With Constant & Trend	t-Statistic	.1900	.0967	.1812	.1605**	.1080	.0671	.1054	
At First Difference									
		d(LITERACY)	d(EDUEXPEN)	d(AYS)	d(TI)	d(CO ₂)	d(GEG)	d(R_D)	
With Constant	t-Statistic	.6132**	.1122	.5605**	.3271	.1543	.1517	.0652	
With Constant & Trend	t-Statistic	.1369	.1128	.1651	.0553	.1381	.1114	.0615	
At Level									
		REC	NREC	TRADE	POPG	URB	GDPPC	INF	EMP
With Constant	t-Statistic	.6495**	.5953**	.2807	.6545**	.7295**	.7021**	.2563	.7308**
With Constant & Trend	t-Statistic	.1204*	.1621**	.1710**	.1618**	.1617**	.1948**	.0860	.0895
At First Difference									
		d(REC)	d(NREC)	d(TRADE)	d(POPG)	d(URB)	d(GDPPC)	d(INF)	d(EMP)
With Constant	t-Statistic	.2243	.3559	.3239	.2424	.4047	.6025**	.0690	.2663
With Constant & Trend	t-Statistic	.1329*	.1589**	.0721	.1427*	.1726**	.1299	.0696	.0983

Note: (*) Significant at the 10%; (**) Significant at the 5%; and (***) Significant at the 1%.

percentage of GDP, and 1 year increase in average year of schooling, will lead to an increase in the *GEG* by .734, .717, and 5.298 index of EAMFP, respectively. However, the levels of national education variables that are effective in increasing *GEG* are 84% literacy rate, 11.9% education expenditure as a percentage of GDP, and 5.5 years of average year of schooling. This means that national education through literacy rate, education expenditure, and average year of schooling are aiding in achieving sustainable development through green economy in China and the cost-effective levels to promote it are 84% literacy rate, 11.9% education expenditure as a percentage of GDP, and 5.5 years average year of schooling. On the other side, *TI* is positively related to the *GEG* but the impact is insignificant, which means it is minimal since it satisfies the a priori expectation of the direction of the relationship where the coefficient is that a 1% increase in *TI*, induces an increase in the *GEG* by .149 index of EAMFP and the level of *TI* on innovations in environment-related technologies effective to promote the *GEG* is 10.4% of all technologies. This suggests that *TI* on innovations in environment-related technologies is promoting *GEG* in China, and the cost-effective level of promoting *GEG* is 10.4% of innovations in environment-related technologies as a percentage of all technologies.

The remaining variables, namely *R&D*, *TRADE*, *URB*, *GDPPC*, *INF*, and *EMP* are positively related to the *GEG*, while *REC*, *NREC*, and *POPG* are negatively related to it. All are significant except the *R&D*, *TRADE* and *GDPPC*; hence, their impacts are minimal, though *R&D* satisfied the a priori expectation that an increase in *R&D* expenditure is promoting the *GEG*. It is evident that a 1% increase in *R&D* as a percentage of GDP, 1% increase in *TRADE* as a percentage of GDP, *URB* as a percentage of total population, 1% increase in *GDPPC*, *INF*, and *EMP* will lead to an increase in *GEG* by 6.916, .038, 3.094, .0003, .088, and 2.004 index of EAMFP, respectively. While a 1% increase in *REC* and *NREC* as a percentage of total energy consumption, 1% increase in *POPG* will lead to a decrease in the green economic growth by .79, 1.876, and 11.144 index of EAMFP, respectively.

Fig. 3 displays the plot of actual vs. predicted for model 2. It can be observed from the figure that the variables used to estimate the effect of education and technological innovation on green economic growth and the policy threshold levels of national education and technological innovation capable of fostering the green economic growth can potentially predict the model.

Table 6 displays the threshold instrumental variables (2SLS)

Table 4

Threshold instrumental variables (2SLS) regression result for model 1.

$$aCO_2 = \beta_0 + \beta_1 EDU + \beta_2 TI + \beta_3 TI^2 + \beta_4 R\&D + \beta_5 NREC + \beta_6 REC + \beta_7 TRADE + \beta_8 URB + \beta_9 POPG + \beta_{10} GDPPC + \beta_{11} INF + \beta_{12} EMP + \mu_t$$

Variables	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
<i>LITERACY</i>	−.231	.073	−3.18	.001	−.089	−.374	b
<i>EDUEXPEN</i>	−.013	.022	−.57	.572	−.056	.031	
<i>AYS</i>	−6.31	2.098	−3.01	.003	−10.422	−2.198	b
<i>TI</i>	.134	.073	1.84	.065	−.009	.277	d
<i>TI</i> ²	−.009	.005	−1.95	.051	−.018	0	d
<i>R&D</i>	2.216	.43	5.15	0	1.373	3.059	b
<i>REC</i>	−.117	.033	−3.51	0	−.182	−.051	b
<i>NREC</i>	.029	.058	.50	.615	−.084	.142	
<i>TRADE</i>	.018	.007	2.77	.006	.005	.031	b
<i>POPG</i>	.261	.371	.71	.481	−.465	.988	
<i>URB</i>	−.321	.142	−2.27	.023	−.598	−.044	c
<i>GDPPC</i>	.003	.004	8.71	0	0	0	b
<i>INF</i>	−.009	.004	−2.08	.038	−.017	0	c
<i>EMP</i>	−.192	.112	−1.72	.085	−.411	.027	d
Constant	38.608	11.864	3.25	.001	15.356	61.86	b
Mean dependent var		4.655	SD dependent var			2.182	
R-squared		.971					
Chi-square		28,279.930	Prob > chi2			.000	
Threshold for <i>LITERACY</i> 93% out of 100							
Threshold for <i>EDUEXPEN</i> 12% of GDP							
Threshold for <i>AYS</i> 6years							
Threshold for <i>TI</i> 10.16% of total							

^a Note that, national education level (*EDU*) following Xin et al. (2023), was composed into three different factors, namely literacy rate (*LITERACY*), education expenditure (*EDUEXPEN*), and average year of schooling (*AYS*).

^b $p < 0.01$.

^c $p < 0.05$.

^d $p < 0.1$.

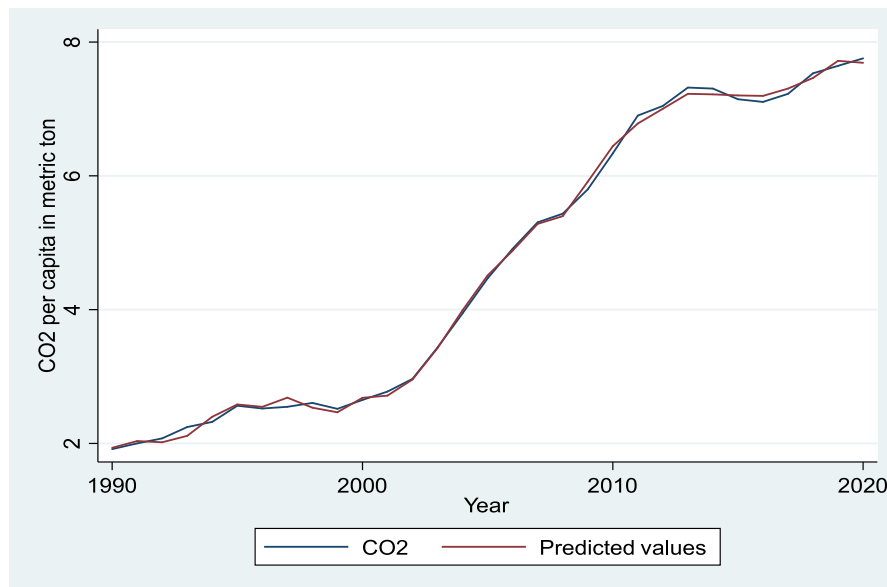


Fig. 2. Plot of actual vs. predicted for model 1.

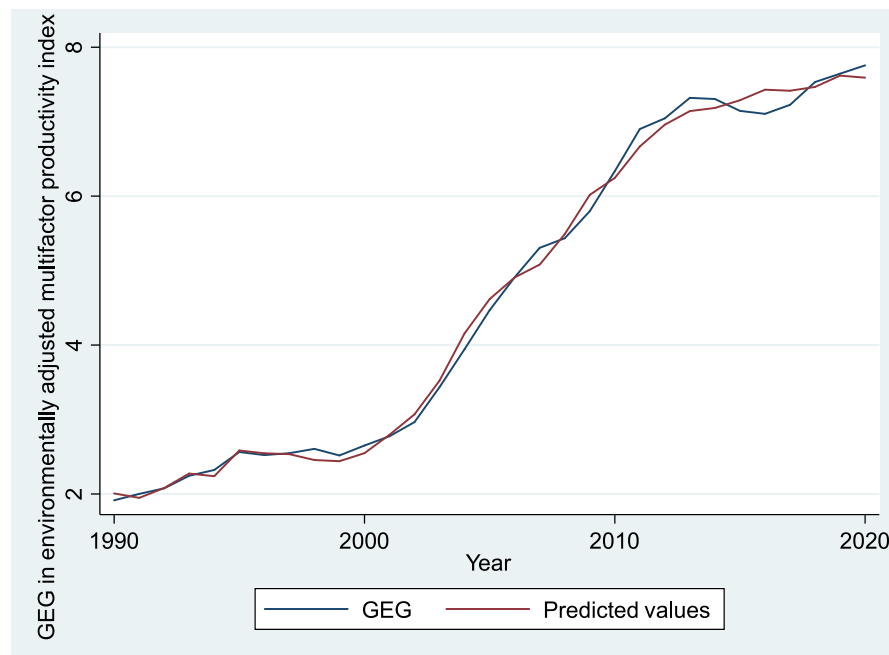
regression results for model 3, which evaluates the interaction effects between national education and technological innovation on CO₂ emissions intensity reduction in China. Therefore, the aim is to ascertain whether national education through innovations in environment-related technologies is reducing the release of CO₂ emissions intensity. According to the table, the interaction terms, namely *EDUEXPEN*TI*, *AYS*TI*, and *LITERACY*TI*² are negatively related to the CO₂, nonetheless, *EDUEXPEN*TI* is not significant even though it also satisfies the a priori expectation of having a negative relationship with the CO₂, hence, it is also impacting the reduction of the emission but weakly. The results show that a 1% increase in literacy rate, 1% increase in education

expenditure as a percentage of GDP, and 1 year increase in average year of schooling, will lead to a decrease in the release of CO₂ emissions intensity through innovations in environment-related technologies by .016, .759, and .003 metric ton of CO₂, respectively. While *LITERACY*TI*, *EDUEXPEN*TI*², and *AYS*TI*² are positively related to the CO₂ and are significant except *EDUEXPEN*TI*². Nevertheless, this is not in line with the a priori expectation of having a negative relationship with the CO₂, which is the theoretical expectation of the relationship. The finding indicates that national education via literacy rate, education expenditure, and average year of schooling through innovations in environment related technologies are assisting in achieving the goal of

Table 5

Threshold instrumental variables (2SLS) regression result for model 2.

$GEG = \beta_0 + \beta_1 EDU + \beta_2 TI + \beta_3 R\&D + \beta_4 NREC + \beta_5 REC + \beta_6 TRADE + \beta_7 URB + \beta_8 POPG + \beta_9 GDPPC + \beta_{10} INF + \beta_{11} EMP + \mu_t$							
Variables	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
LITERACY	.734	.747	.98	.326	.731	2.199	
EDUEXPEN	.717	.229	3.13	.002	1.165	.268	a
AYS	5.298	2.401	2.47	.014	6.203	7.394	b
TI	.149	.233	.64	.524	-.309	.606	
R&D	6.916	4.518	1.53	.126	-1.938	15.771	
REC	-.79	.326	-2.42	.015	-1.428	-.151	b
NREC	-1.876	.553	-3.39	.001	-2.96	-.793	a
TRADE	.038	.071	.53	.595	-.101	.176	
POPG	-11.144	3.685	-3.02	.002	-11.366	-13.921	a
URB	3.094	1.511	2.05	.041	.132	6.056	b
GDPPC	.0003	.0004	.76	.446	.0002	.0004	
INF	.088	.044	1.98	.047	.001	.175	b
EMP	2.004	1.177	1.70	.089	-.302	4.31	c
Constant	-11.87014	10.6	1.45	.147	-63.374	425.191	
Mean dependent var		1.959	SD dependent var			2.073	
R-squared		.862					
Chi-square		193.143	Prob > chi2			.000	
Threshold for LITERACY 84% out of 100							
Threshold for EDUEXPEN 11.9% of GDP							
Threshold for AYS 5.5years							
Threshold for TI 10.4% of total							

^a $p < 0.01$.^b $p < 0.05$.^c $p < 0.1$.**Fig. 3.** Plot of actual vs. predicted for model 2.

reducing the release of CO₂ emissions intensity in China. Hence, proven the assertion of Gu et al. (2019) on identifying the possibility of an inverted U-shaped EKC between *TI*, specifically innovations in environment-related technologies, and CO₂ emissions been the quadratic coefficient is negative and significant.

However, the remaining variables, namely *REC*, *URB*, *INF*, and *EMP*, are negatively related with CO₂, while *R&D*, *NREC*, *TRADE*, *POPG*, and *GDPPC* are positively related with the CO₂. All are significant except the *NREC*, *POPG*, and *EMP*, hence, their impacts are minimal though satisfied the a priori expectation that increase in nonrenewable energy consumption, population growth, and GDP per capita could increase the release of CO₂ emissions intensity. Thus, a 1% increase in *REC* as a

percentage of total energy consumption, *URB* as a percentage of total population, *INF*, and *EMP* will lead to a decrease in CO₂ emissions intensity by .161, .597, .008, and .165 metric ton of CO₂, respectively. Whereas, a 1% increase in *R&D* as a percentage of GDP, *NREC* as a percentage of total energy consumption, 1% increase in *TRADE* as a percentage of GDP, 1% increase in *POPG*, and a 1% increase in GDP per capita will lead to an increase in the release of the CO₂ emissions intensity by 2.35, .035, .013, .128, and .00003 metric ton of CO₂, respectively.

Fig. 4 depicts the plot of actual vs. predicted for model 3. It can be observed from the figure that the variables used to evaluate the interaction effects between national education and technological innovation

Table 6
Threshold instrumental variables (2SLS) regression result for model 3.

$$CO_2 = \beta_0 + \beta_1 EDU*TI + \beta_2 EDU*TI^2 + \beta_3 R\&D + \beta_4 NREC + \beta_5 REC + \beta_6 TRADE + \beta_7 URB + \beta_8 POPG + \beta_9 GDPPC + \beta_{10} INF + \beta_{11} EMP + \mu_t$$

Variables	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
<i>LITERACY*TI</i>	.054	.018	3.07	.002	.02	.089	a
<i>EDUEXPEN*TI</i>	−.016	.013	−1.20	.231	−.042	.01	
<i>AYS*TI</i>	−.759	.261	−2.91	.004	−1.27	−.247	a
<i>LITERACY*TI</i> ²	−.003	.001	−3.31	.001	−.005	−.001	a
<i>EDUEXPEN*TI</i> ²	.002	.002	1.35	.177	−.001	.005	
<i>AYS*TI</i> ²	.044	.014	3.18	.001	.017	.071	a
<i>R&D</i>	2.35	.452	5.19	0	1.463	3.237	a
<i>REC</i>	−.161	.028	−5.82	0	−.216	−.107	a
<i>NREC</i>	.035	.065	.54	.592	−.093	.163	
<i>TRADE</i>	.013	.007	1.92	.055	.0003	.027	c
<i>POPG</i>	.128	.374	.34	.733	−.605	.861	
<i>URB</i>	−.597	.078	−7.70	0	−.749	−.445	a
<i>GDPPC</i>	.00003	.000004	7.49	0	.00002	.00041	a
<i>INF</i>	−.008	.004	−1.86	.063	−.017	.00042	c
<i>EMP</i>	−.165	.114	−1.45	.148	−.388	.058	
Constant	30.299	12.39	2.45	.014	6.015	54.582	b
Mean dependent var		4.655				2.182	
R-squared		.999					
Chi-square		27,786.481	Prob > chi2			.000	

^a $p < 0.01$.

^b $p < 0.05$.

^c $p < 0.1$.

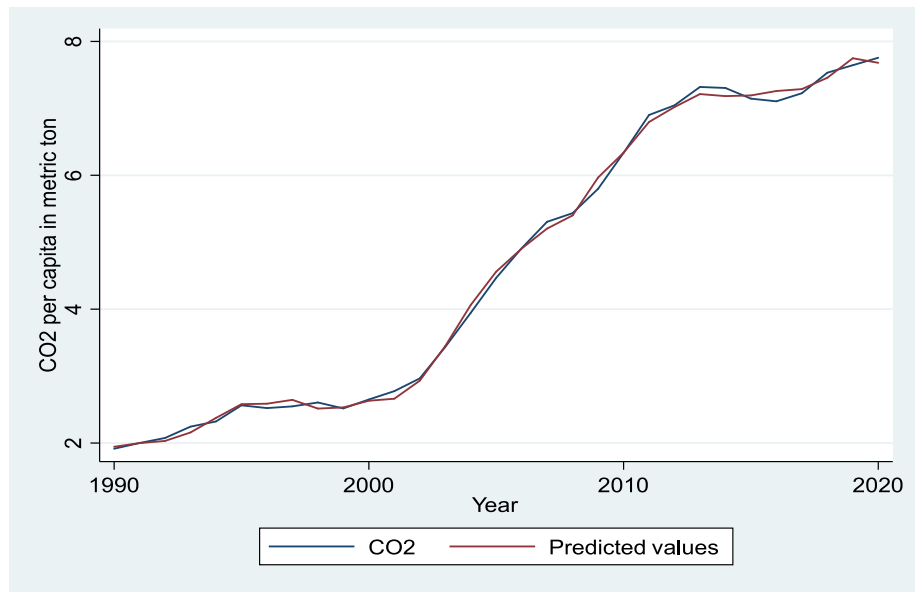


Fig. 4. Plot of actual vs. predicted for model 3.

on CO₂ emissions intensity reduction, identifying synergies and trade-offs can potentially predict the model.

Table 7 shows the threshold instrumental variables (2SLS) regression outcomes for model 4, which scrutinizes the interaction effects of national education and technological innovation on green economic growth in China. Therefore, it aims to determine whether national education through innovations in environment-related technologies is encouraging green economic growth. In line with the table, the interaction terms, namely *LITERACY*TI* and *EDUEXPEN*TI* are negatively related with the *GEG* but *LITERACY*TI* is insignificant; however, both are not in line with the a priori expectation of having a positive relationship with the *GEG*, which is the theoretical expectation of the relationship. *AYS*TI* is positively related with the *GEG*, but the impact is not significant even though it satisfies the a priori expectation of having a positive relationship with the *GEG*; hence, it is promoting the growth of the green economy, though the impact is weak. It shows that a 1-year

increase in average year of schooling, will lead to an increase in the growth of the green economy by .491 index of EAMFP.

The finding designates that national education through innovations in environment-related technologies is weakly encouraging the *GEG* in China. Regarding the other variables, *R&D*, *GDPPC*, *INF*, and *EMP* are positively related with the *GEG* but none is significant; however, they satisfied the a priori expectations except the *INF*, where a 1% increase in *R&D* as a percentage of GDP, *GDPPC*, and *EMP* will lead increase of 3.756, .000098, and 1.707 index of EAMFP for *GEG*, respectively. While *REC*, *NREC*, *TRADE*, *POPG*, and *URB* are positively related with the *GEG* but *TRADE* and *URB* are insignificant and not in line with the a priori expectation of expecting to have negative relationship with the *GEG* been increase in these variables is theoretically considered as invitation for increase in the release of CO₂ emissions intensity, just as in *NREC*, *TRADE*, and *POPG* even though they are significant.

Therefore, only those that are either significant or in line with the a

Table 7

Threshold instrumental variables (2SLS) regression result for model 4.

$GEG = \beta_0 + \beta_1 EDU * TI + \beta_3 R \& D + \beta_4 NREC + \beta_5 REC + \beta_6 TRADE + \beta_7 URB + \beta_8 POPG + \beta_9 GDPPC + \beta_{10} INF + \beta_{11} EMP + \mu_t$							
Variables	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
LITERACY*TI	-.028	.051	-.56	.578	-.129	.072	
EDUEXPEN*TI	-.048	.025	-1.96	.05	-.097	0	b
AYS*TI	.491	.78	.63	.529	-1.038	2.02	
R&D	3.756	4.9	.77	.443	-5.849	13.36	
REC	-1.317	.254	-5.18	0	-1.816	-.819	a
NREC	-1.394	.5	-2.79	.005	-2.374	-.414	a
TRADE	-.045	.077	-.59	.555	-.196	.105	
POPG	-9.738	3.228	-3.02	.003	-16.065	-3.411	a
URB	-.685	.767	-.89	.371	-2.188	.817	
GDPPC	.000098	.00038	.26	.797	-.001	.001	
INF	.057	.05	1.13	.259	-.042	.155	
EMP	1.707	1.053	1.62	.105	-.357	3.772	
Constant	55.978	94.639	.59	.554	-129.511	241.467	
Mean dependent var		1.959	SD dependent var			2.073	
R-squared		.821					
Chi-square		142.275	Prob > chi2			.000	

** $p < 0.05$.^a $p < 0.01$.^b $p < 0.1$.

priori expectation or both should be given attention to. Hence, *REC*, *NREC*, and *URB* where a 1% increase in *REC* and *NREC* as a percentage of total energy consumption, and a 1% increase in *URB* as a percentage of total population, will induce a decrease in *GEG* by 1.317, 1.394, and .685 index of EAMFP.

Fig. 5 displays the plot of actual vs. predicted for model 4. It is evident from the figure that the variables used to evaluate the interaction effects between national education and technological innovation on green economic growth, identifying synergies and trade-offs have the element for predicting the model.

5. Discussions

China's rapid industrialization and economic growth have led to significant increases in CO₂ emissions, creating an urgent need for effective strategies to achieve environmental sustainability. Education and TI are crucial factors in this context, as they can drive reductions in CO₂ emissions and foster GEG. Prior research suggests that higher

education levels enhance environmental awareness and skills, leading to more sustainable practices, while advancements in TI contribute to cleaner production processes and energy efficiency. The findings of this study reveal that education is instrumental in lowering the degree of intensity of CO₂ emissions. This finding is backed by Yao et al. (2019), who suggested that pollution emissions are diminished by improving human capital. The findings indicate that pollution emissions decrease as a consequence of an increase in education. The empirical findings suggest that humans are essential in the maintenance of environmental quality, particularly in the Chinese economy. Furthermore, education is considered the most critical determinant in developed countries. Human capital mitigates renewable energy scarcity by restricting CO₂ emissions. The findings also implies that China's educational systems and fiscal expenditures are conducive to environmental quality. It is also consistent with the findings of Xin et al. (2023) that the literacy rate and average year of schooling are contributing to the reduction of CO₂ emissions. Moreover, it supports the conclusions of Akram et al. (2019) and Shahbaz et al. (2019) that education has a detrimental impact on

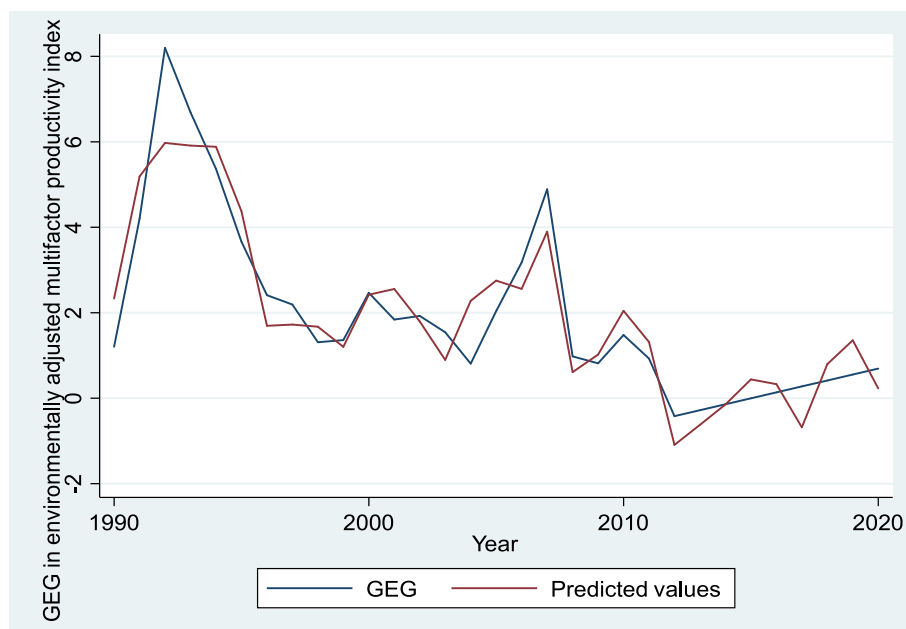


Fig. 5. Plot of actual vs. predicted for model 4.

energy consumption, which in turn results in an improvement in environmental quality. In the same vein, [Li and Ouyang \(2019\)](#) have shown that increased human capital leads to both an increase in short-term CO₂ emissions and a decrease in long-term CO₂ emissions.

Similarly, it was found that education is helping to achieve GEG. This is consistent with [Zhang et al. \(2023\)](#) findings, which show that improving educational quality can help students get the variety of skills needed for green jobs. Therefore, the increase in the number of people with green skills will facilitate the provision of green employment, which will in turn lead to an increase in the activities of green growth in society. In a similar vein, [Zhang and Shang \(2022\)](#) have shown that education plays a significant role in GEG by fostering regional green development, fostering economic expansion, and protecting the environment. According to [Deshuai et al. \(2023\)](#), education serves as a bridge between higher education and GEG by facilitating knowledge generation and transformation, which promotes TI. [Albanese et al. \(2022\)](#) argued that education plays an important role in the development of a green economy, as indicated by the fact that green GDP is more adaptable to changes in higher education than traditional GDP. [Gao et al. \(2019\)](#) found that policies promoting green education in higher education assist human capital become more environmentally conscientious. They also have an impact on labour skills and green culture, both of which are important for the future low-carbon economy. Moreover, though firm-level green subsidies have a greater impact on short-term low-carbon productivity, it is necessary to incorporate a green education stimulus to improve the quality of green education and support sustainable growth in the long run.

The results also showed that environmental technology advancements are reducing CO₂ emissions and promoting GEG. This is in line with the findings made by [Liu et al. \(2021\)](#) that technical innovation in energy exchange can successfully lower CO₂ emissions. Furthermore, studies by [Cheng et al. \(2021\)a,b](#), [Khalid et al. \(2018\)](#), and [Zheng \(2021\)](#) have shown that TI can improve energy efficiency and notably reduce CO₂ emissions. Initially, the sharing economy enhances the TI of enterprises through the sharing of knowledge, which is a critical method for reducing CO₂ emissions and improving energy efficiency. Moreover, [Weng et al. \(2018\)](#) have contributed to the advancement of energy internet technologies, which are distinguished by the comprehensive incorporation of information technology and new energy technologies. These technologies are widely regarded as the solution to the issues of environmental pollution and fossil energy scarcity. In the same vein, [Lee and Mo \(2011\)](#) have demonstrated that environmental innovation in biofuel is linked to a reduction in CO₂ emissions, without any economic implications. [Carrion-Flores and Innes \(2010\)](#) have also found a significant decrease in toxic pollution as a result of the implementation of environmental innovation and stringent policy. However, CO₂ emissions may not be significantly reduced through the implementation of environmental innovation if the time horizon is limited. Therefore, the advantages of implementing environmental innovation are contingent upon their long-term implementation ([Mongo et al., 2021](#)). Besides, [Wang and Zhu \(2020\)](#) emphasized that renewable energy is more effective in reducing CO₂ emissions than fossil fuel energy in promoting environmental innovation. However, the utilization of the interaction term in the nexus between TI and CO₂ emissions demonstrated the potential for an inverted U-shaped EKC between TI, specifically innovations in environment-related technologies, and CO₂ emissions, as interpreted by [Gu et al. \(2019\)](#).

6. Concluding remarks

Numerous nations, spanning developed, developing, and underdeveloped regions, strive to identify more effective and intelligent strategies for achieving sustainability. The advancement of environmentally sustainable technologies and the enhancement of education are crucial for attaining ecological preservation. Hence, this study aims to contribute to the existing literature by investigating the relationship

between education and TI in reducing CO₂ emissions and promoting GEG in China through marginal effects and policy threshold analyses for attaining the millennium of sustainable development with a timeframe data over the period of 1990–2020 using advanced econometric methods include the Th-IV2SLS model in which the study links theory with practice in providing empirical facts; thus, its findings could be more effective than the previous ones.

The results revealed that the national education level is limiting the release of CO₂ emissions intensity in China and the cost-effective levels to reduce the release of the emissions are 93% literacy rate, 12% education expenditure as a percentage of GDP, and 6years average year of schooling. Second, innovations in environment-related technologies are also discouraging the rising of the release of CO₂ emissions intensity, and the cost-effective level to reduce the release of the emissions is 10.16% of innovations in environment-related technologies as a percentage of all technologies. Third, with respect to GEG, national education was found to be its promoter and the cost-effective levels to promote it are 84% literacy rate, 11.9% education expenditure as a percentage of GDP, and 5.5years average year of schooling. Fourth, innovations in environment-related technologies were also found to be an influencer of the growth in the green economy, and the cost-effective level to influence it is 10.4%. Fifth, when examining the influence of national education on the innovations in environment-related technologies in curbing the release of CO₂ emissions, the finding revealed and effective synergy between the two in tackling CO₂ emissions; however, the performance of these mechanisms is weakly effective in promoting the GEG. The policy recommendations are provided in line with these findings based on each of the three objectives of the study as follows.

• With respect to education on CO₂ emissions and GEG

Education through literacy rate, education expenditure, and years of schooling are going in the direction of curbing the increase of the spread of CO₂ emissions and fostering GEG. Thus, the direction of the country's education is helpful in reducing the effect of CO₂ emissions and promoting GEG. Therefore, the existing educational measures toward the reduction of the CO₂ emissions and encouraging the GEG should be maintained, with possible re-strengthening to achieve more results, where educators, educational institutions, and government should put more designation on the curricula that extend beyond the dissemination of information but to supplement it with the teaching of the full climate project course as a general study. This encompasses the instruction of environmental conservation, renewable energy, climate change, and sustainable practices. Other measures include changing to energy-saving light bulbs in schools; switching off lights in empty classrooms and school halls; avoiding leaving computers, TVs and printers on standby; unplugging chargers when not in use; and turning the school's heating down a degree or two. Furthermore, there is a need to give more orientation to educate educators on the principles of the green economy and the most effective instructional methods. Moreover, education expenditure should prioritize the green budget in order to reduce CO₂ emissions and promote GEG. This entails the allocation of funds in accordance with environmentally favorable policies. In addition, there is a need to integrate green economy-related skills and knowledge into the curriculum during the year.

• With respect to technological innovation on CO₂ emissions and GEG

TI is going in the right direction in the country in curbing the CO₂ emissions and fostering GEG. Therefore, the country's existing TI through environmental related technologies toward the reduction of the CO₂ emissions and promoting the GEG should be maintained, with possible re-strengthening to achieve more results, where substantial resources should be allocated to research and development in environmental technologies by governments, private corporations, and research

institutions. Furthermore, breakthroughs in sustainable agriculture, waste management, and renewable energy can be expedited through collaborative endeavors among academia, industry, and government. Moreover, encouraging the dissemination of innovative solutions and best practices through collaborative networks and open-source initiatives. Besides, it is imperative to allocate resources to educational and training initiatives that pertain to environmental technologies. This encompasses seminars, university courses, and vocational training.

• **With respect to synergetic effects of education and TI in reducing the spread of CO₂ emissions and fostering GEG**

Considering that the combined effect of education and TI on environmental technologies can significantly reduce CO₂ emissions, policymakers should integrate these strategies to achieve better results in curbing this menace. However, if the combined effect proves weak in achieving GEG, policymakers should not combine these strategies. Instead, they should apply them separately to achieve better GEG, as their separate impacts are more effective. In other words, it is more beneficial for policymakers to implement education and TI independently rather than together when aiming to achieve GEG.

However, the limitation of this study is that CO₂ emissions are utilized as an environmental pollutant measure, while CH₄ and N₂O are disregarded. This decision was made to establish a clear focus and a policy targeting mechanism for effectiveness, as CO₂ emissions are the primary greenhouse gas emitted through human activities (EPA, 2024). In order to acquire a comprehensive perspective on a large number of countries, future research should be conducted on a greater number of countries, particularly those that are among the top emitters or top industrialized countries. Furthermore, future research could investigate sector-specific impacts and regional disparities in accordance with the data's availability.

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CRediT authorship contribution statement

Min He: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Babar Nawaz Abbasi:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Zhi-chao Fan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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